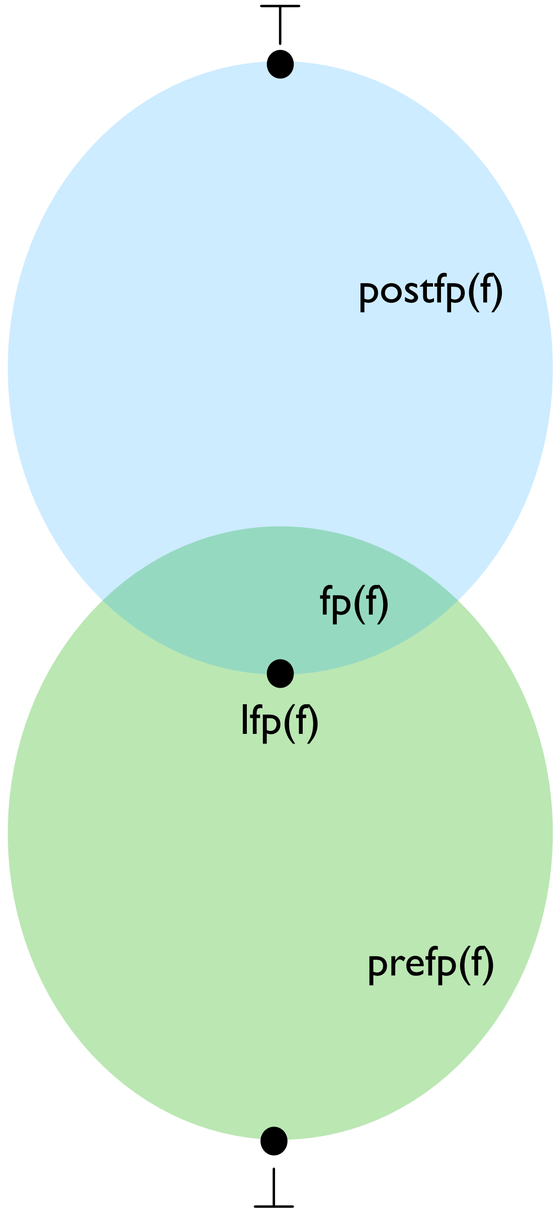
Static analysis and software verification

Lecture 11 - The interval abstract domain

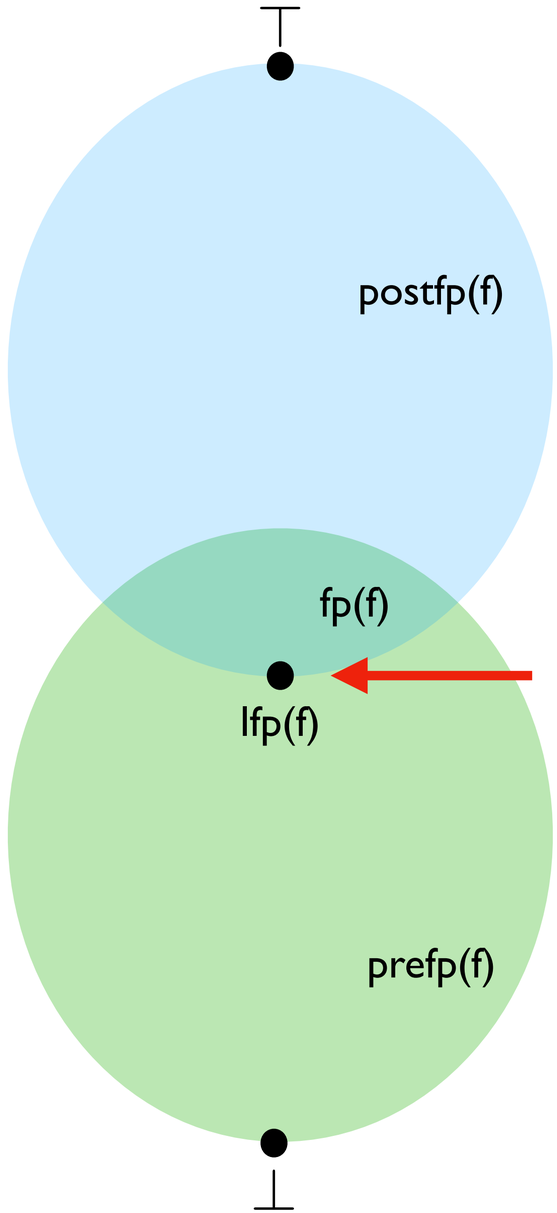
Vincenzo Arceri - University of Parma - vincenzo.arceri@unipr.it

The interval abstract domain

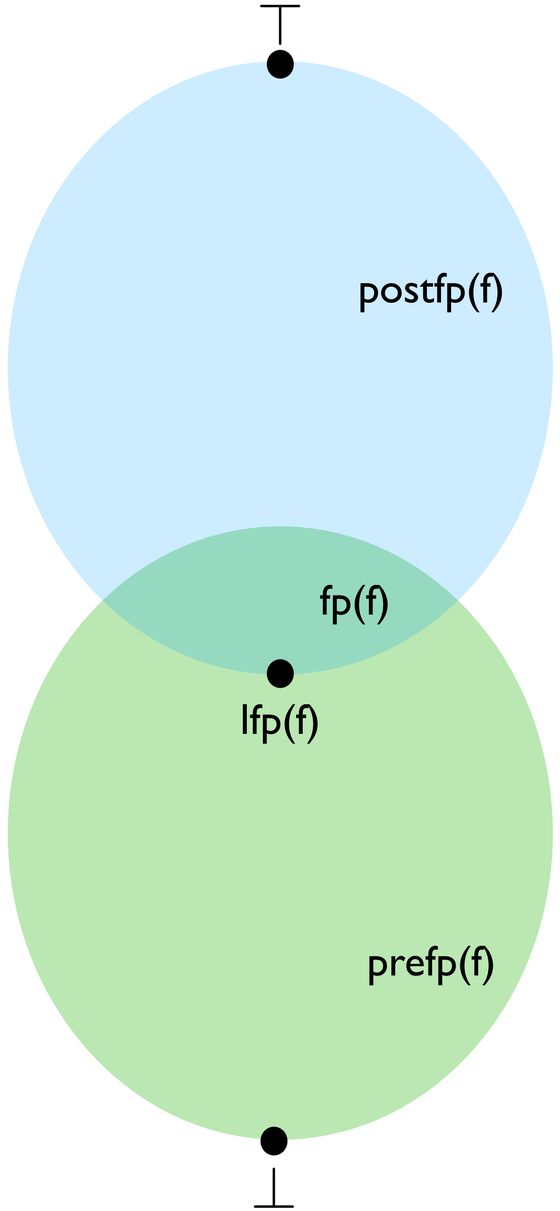
Widening



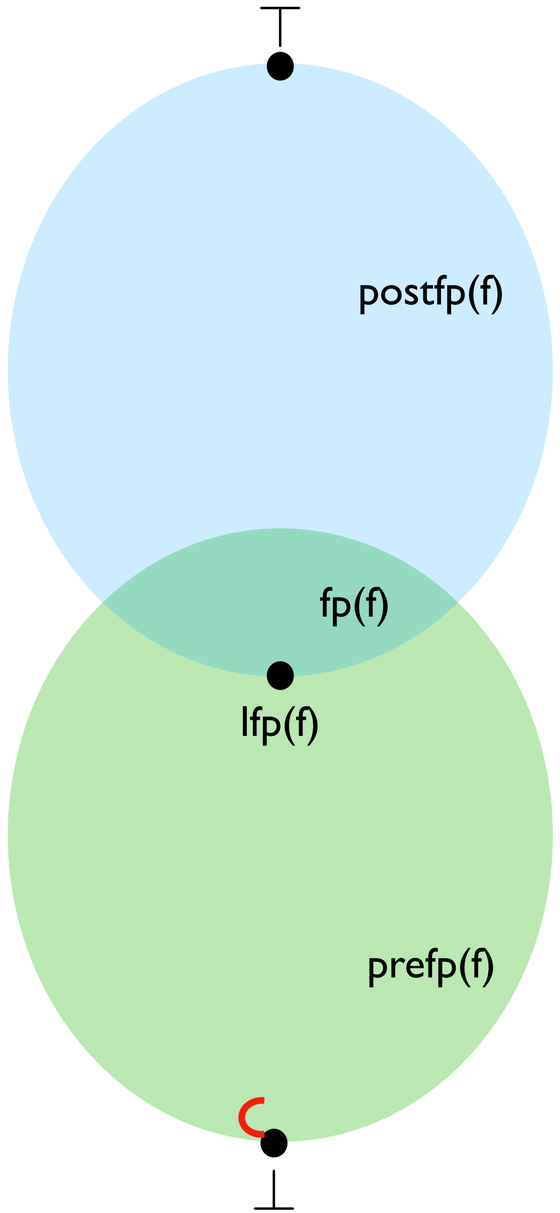
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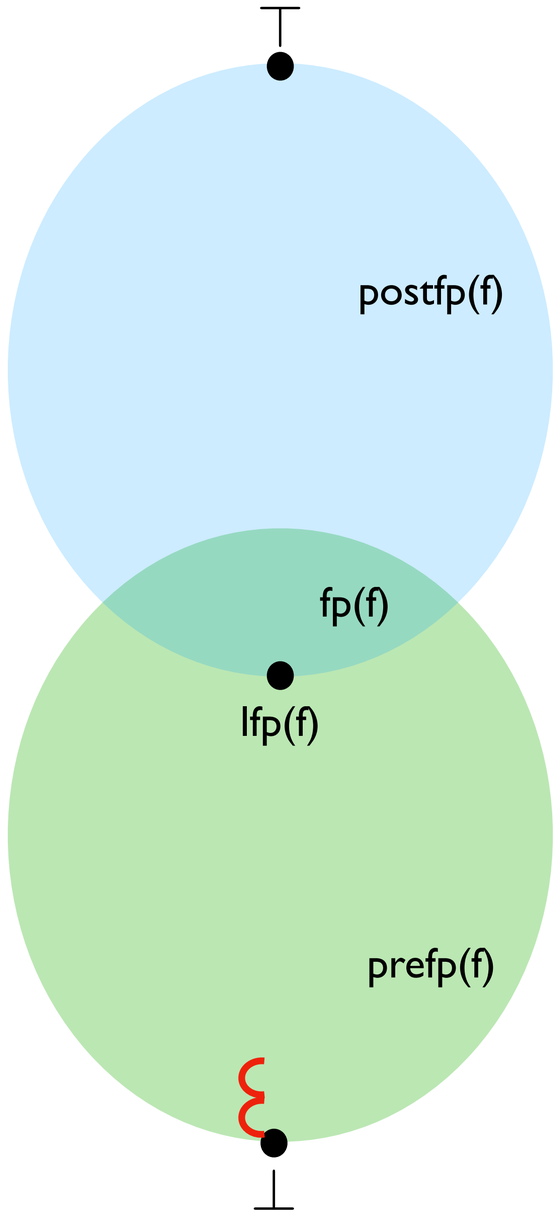
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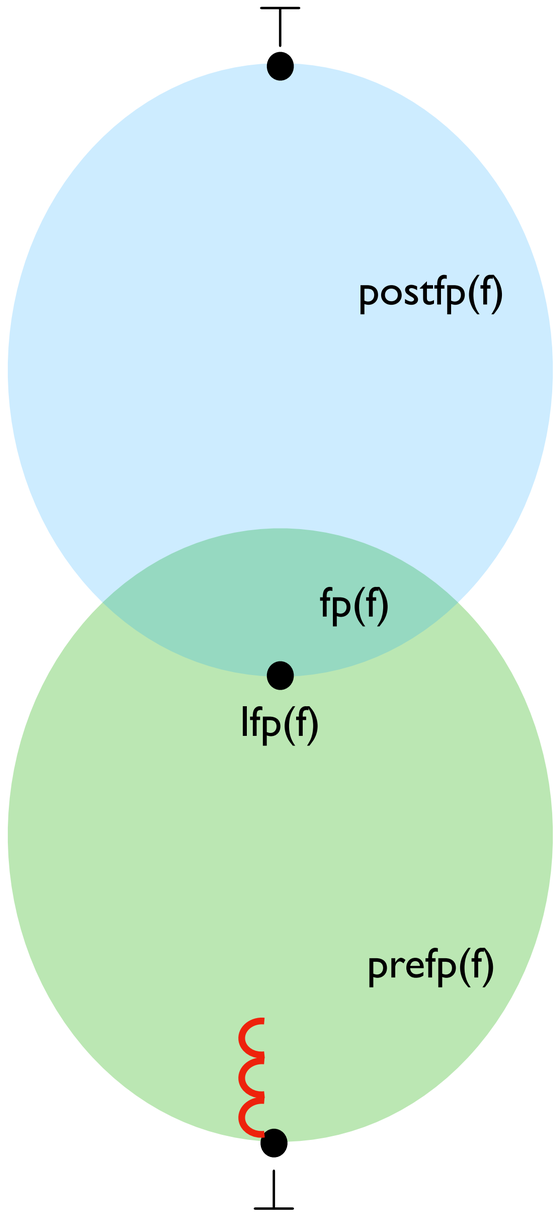
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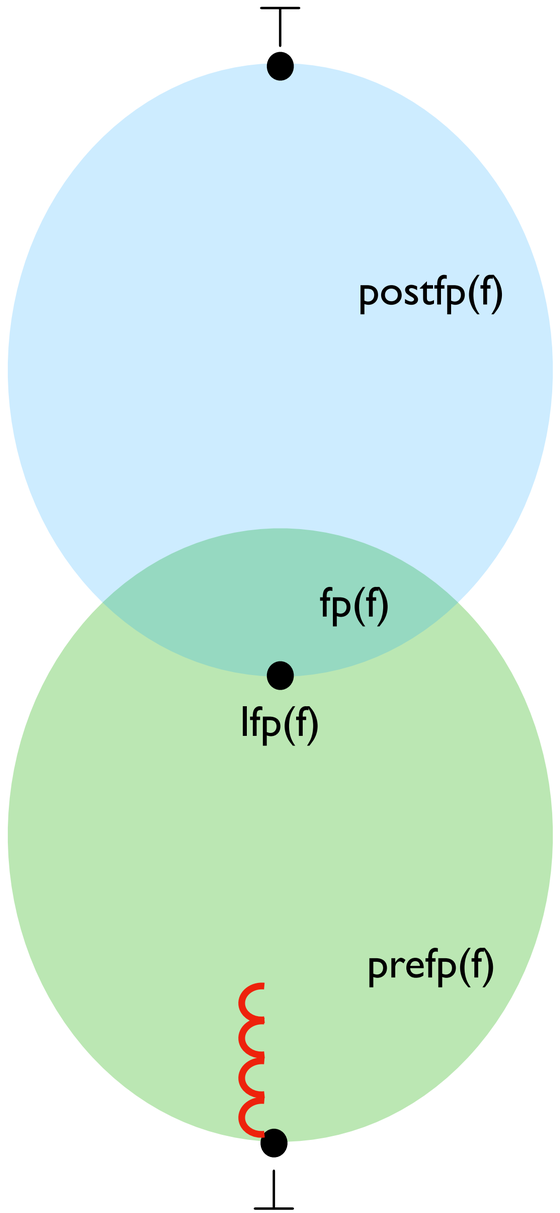
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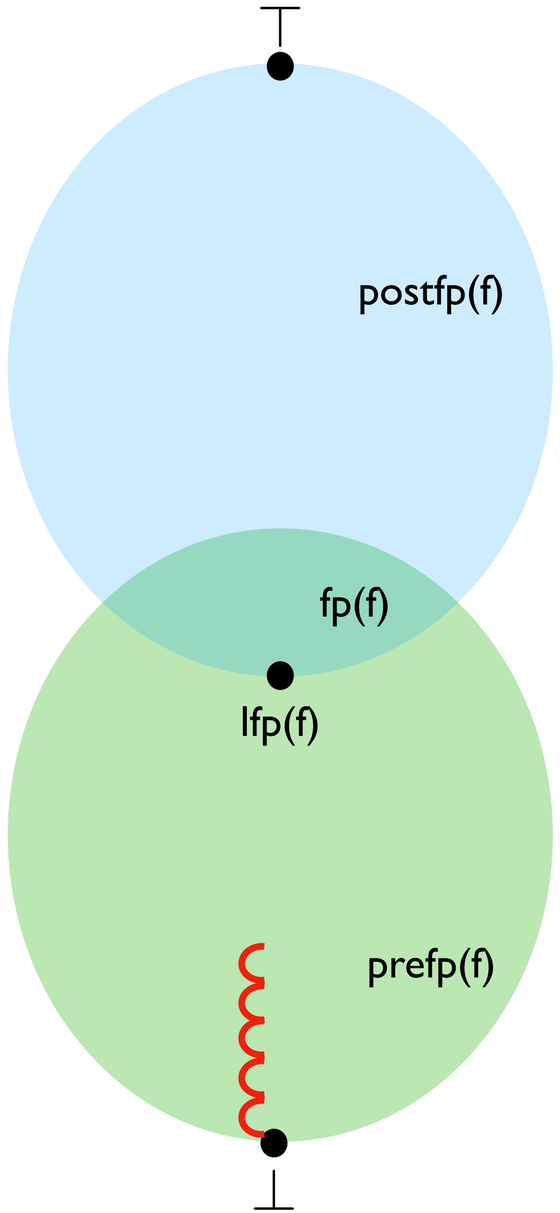
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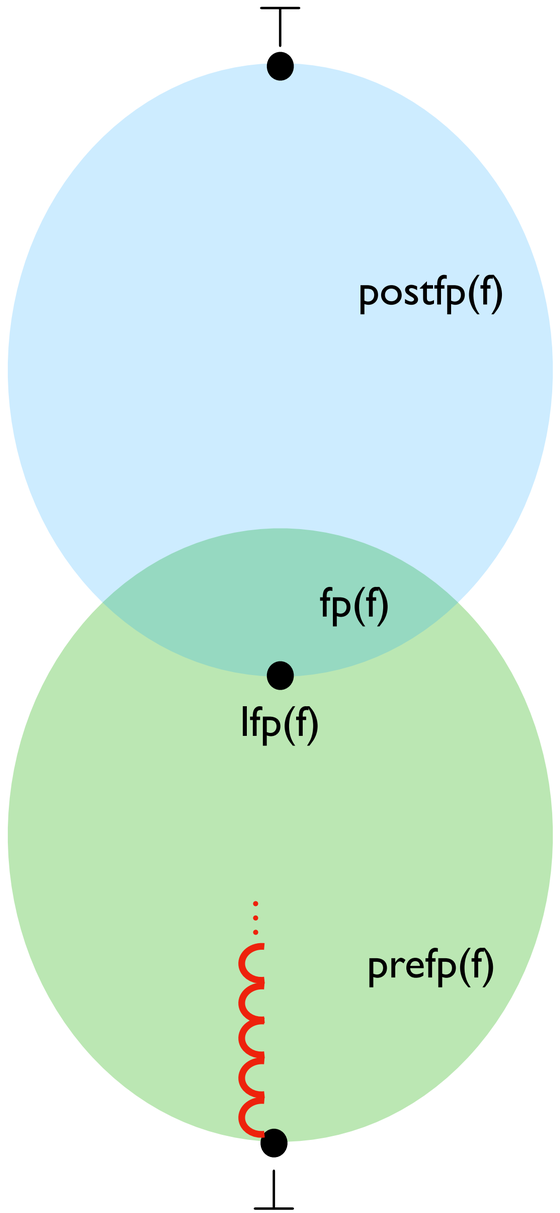
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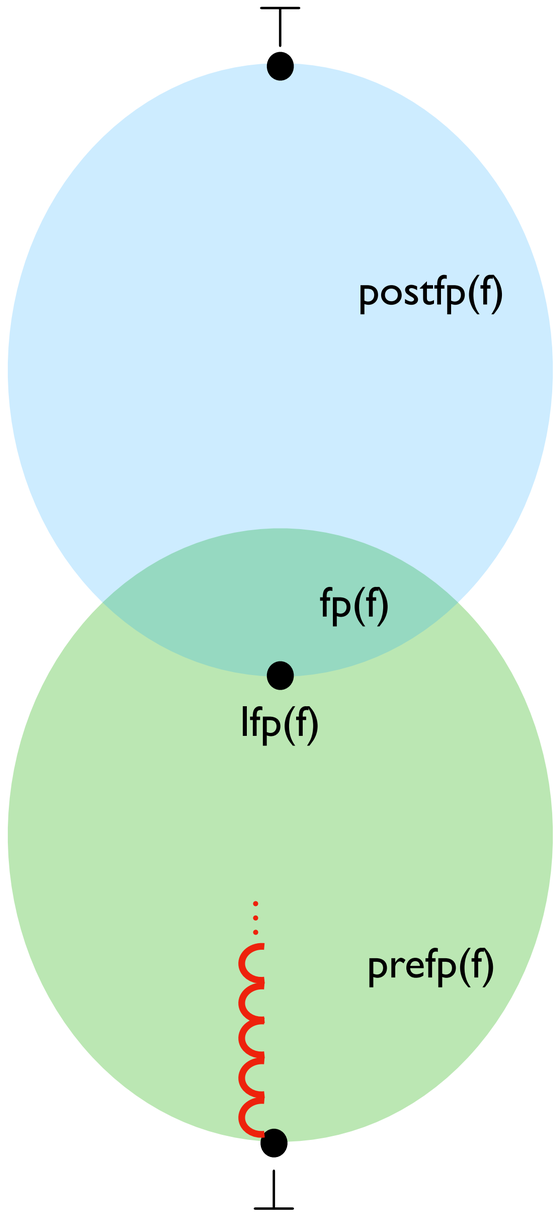
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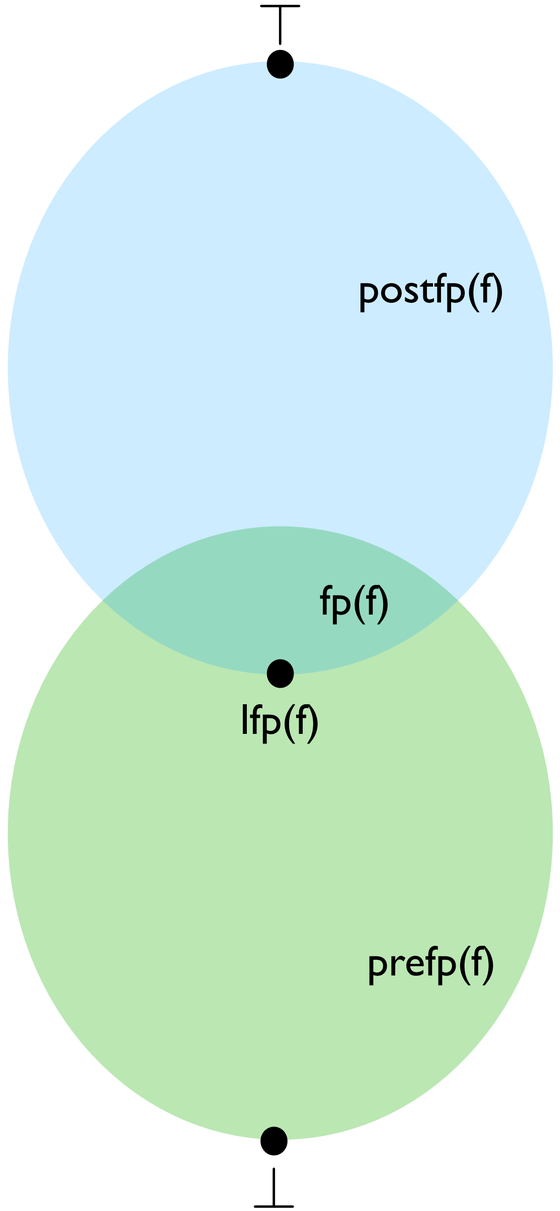


Widening

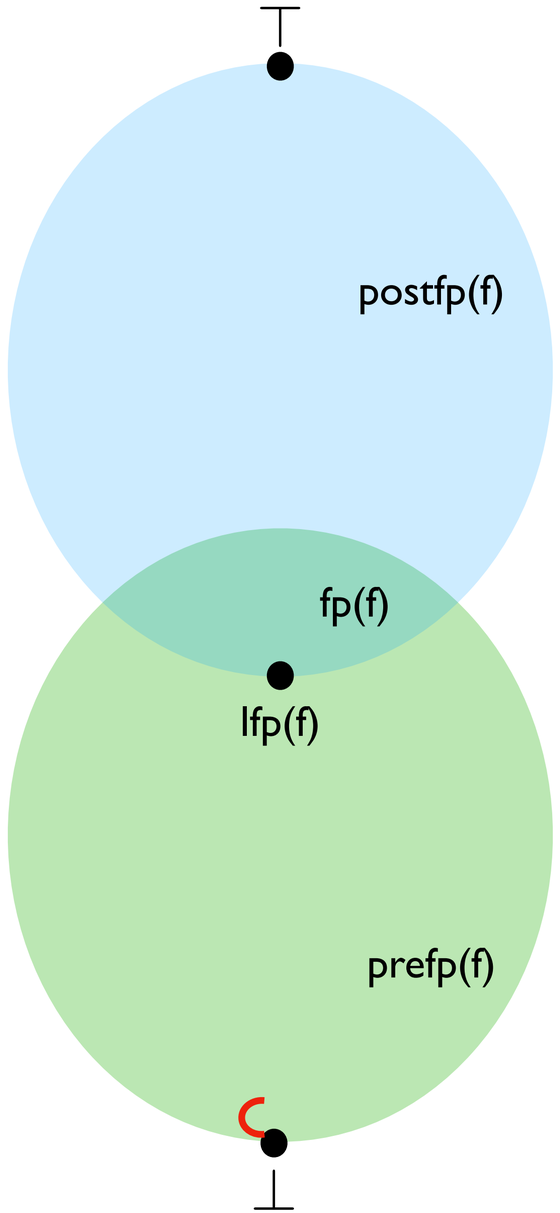


Divergence!

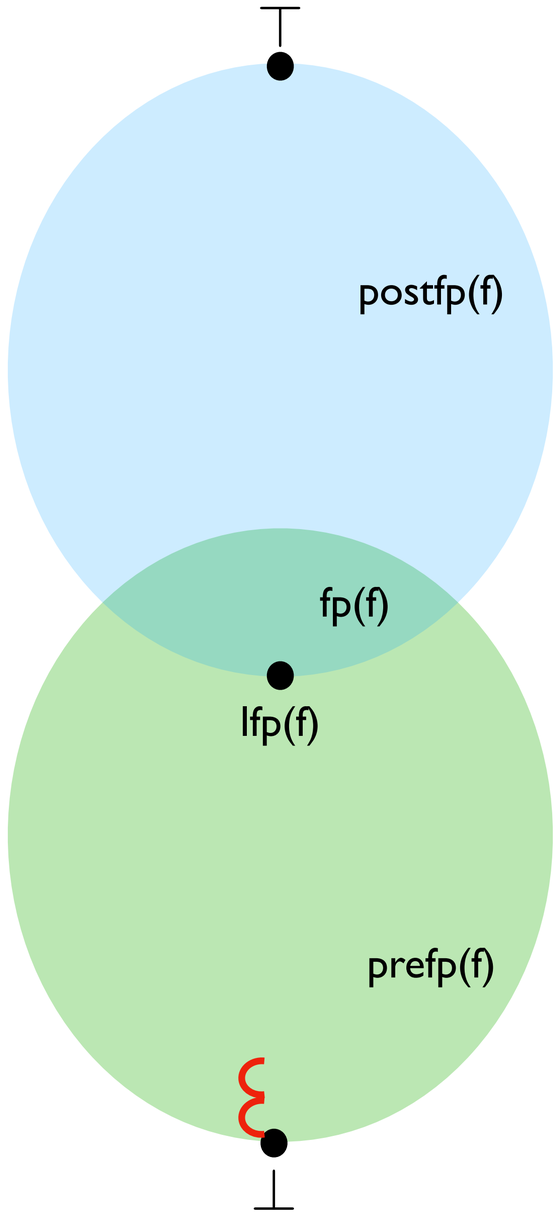
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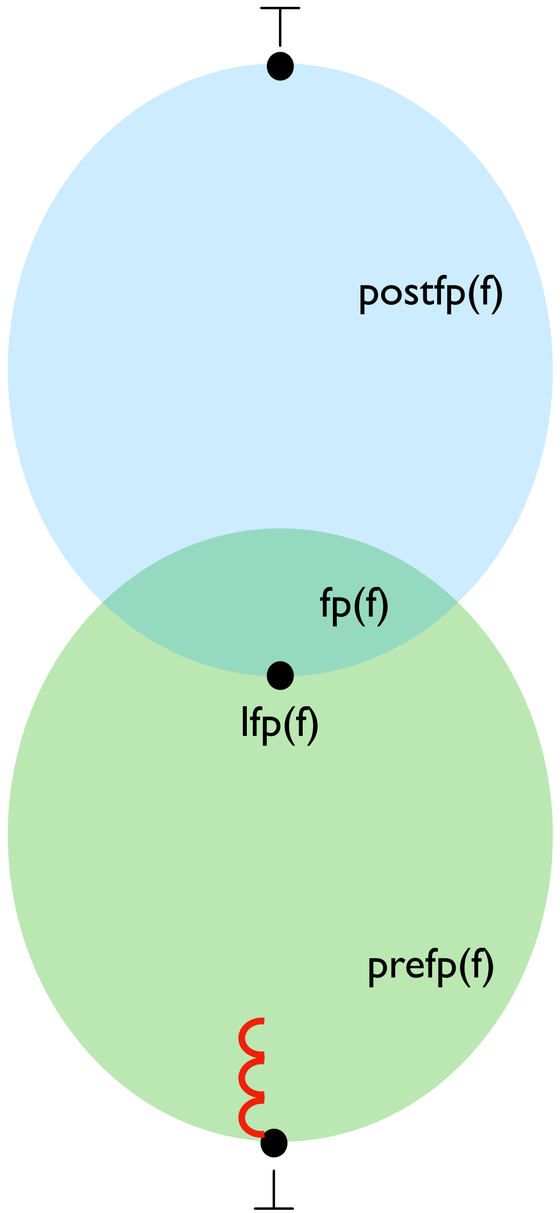
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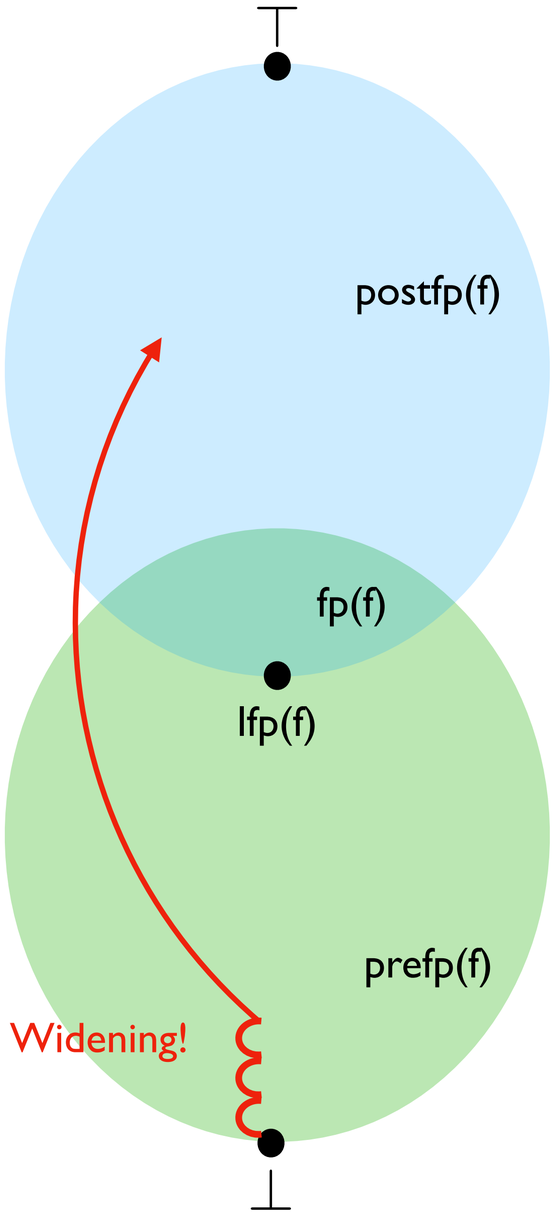
Widening



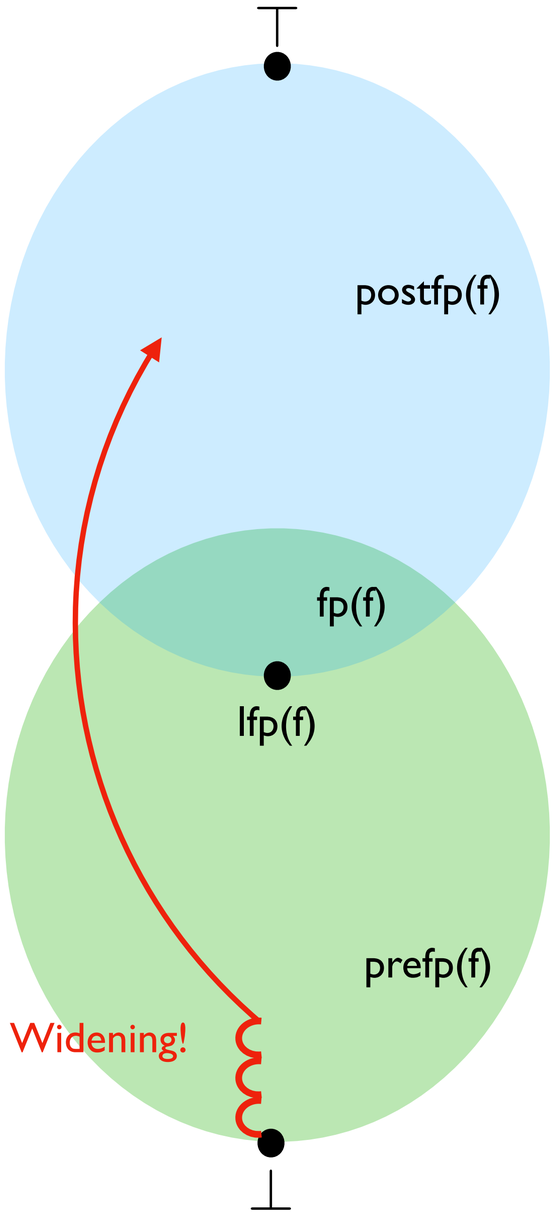
Widening



Widening



Widening



Widening: over-approximates
the lfp(f)
but guarantees convergence

Widening

Widening

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Example of widening?

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$$x \nabla y = \begin{cases} x & \text{if } y \sqsubseteq x \\ \top & \text{otherwise} \end{cases}$$

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Naive

Widening for intervals

$$[a,b]\,\nabla\,[c,d] = \left[\ \begin{cases} a & \text{if } a \le c \\ -\infty & \text{otherwise} \end{cases}, \begin{cases} b & \text{if } b \ge d \\ +\infty & \text{otherwise} \end{cases}\ \right]$$

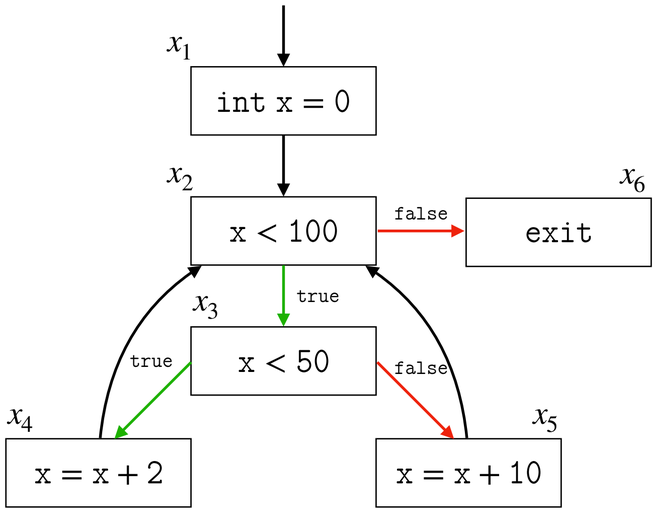
Widening for intervals

$$[a,b]\,\nabla\,[c,d] = \left[\ \begin{cases} a & \text{if } a \le c \\ -\infty & \text{otherwise} \end{cases}, \begin{cases} b & \text{if } b \ge d \\ +\infty & \text{otherwise} \end{cases}\ \right]$$

- Not commutative: $[2,3]\,\nabla\,[1,4] = [-\infty, +\infty], [1,4]\,\nabla\,[2,3] = [1,4]$
- Not monotone: let $[0,1] \sqsubseteq [0,3]$, $[0,1]\,\nabla\,[0,2] = [0, +\infty], [0,3]\,\nabla\,[0,2] = [0,3]$

Widening

Where to apply it?



$$x_1 = \alpha(I)$$

$$x_2 = [[\mathrm{x} = 0]]^\sharp x_1 \sqcup [[\mathrm{x} = \mathrm{x} + 2]]^\sharp x_4 \sqcup [[\mathrm{x} = \mathrm{x} + 10]]^\sharp x_5$$

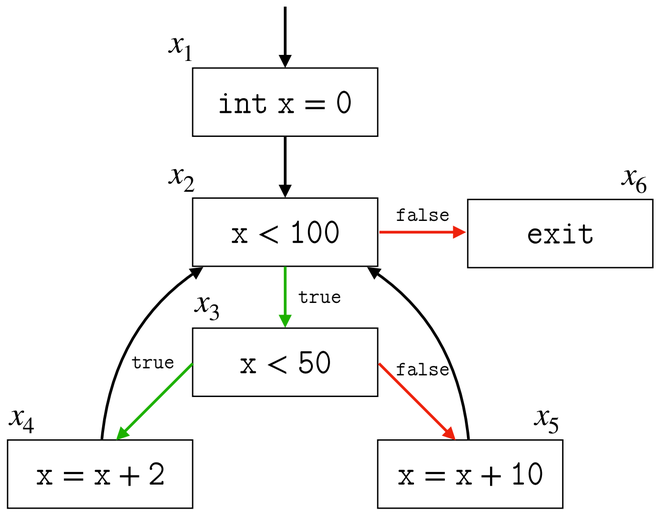
$$x_3 = [[\mathrm{x} < 100]]^\sharp x_2$$

$$x_4 = [[\mathrm{x} < 50]]^\sharp x_3$$

$$x_5 = [[\mathrm{x} >= 50]]^\sharp x_3$$

Widening

Where to apply it?



Without widening

$$F^\sharp = \{x_i = x_i \sqcup f_i^\sharp(x_1, \ldots, x_m) \mid i = 1,2,\ldots,m\}$$

With widening

$$F^\sharp = \{x_i = x_i \nabla f_i^\sharp(x_1, \ldots, x_m) \mid i = 1,2,\ldots,m\}$$

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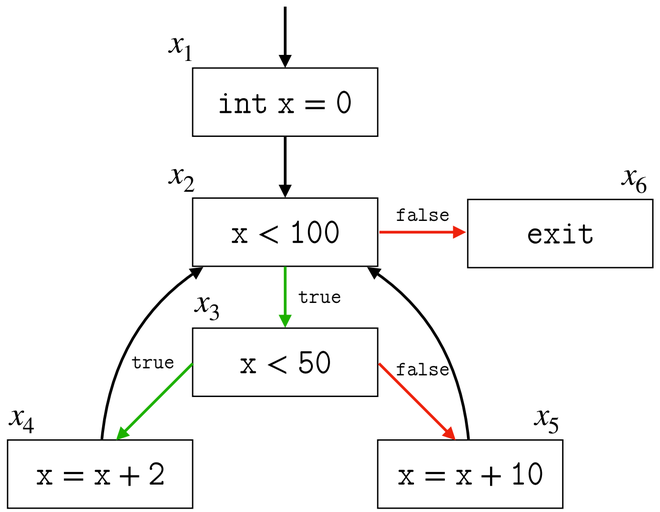
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Widening

Where to apply it?



Without widening

$$F^\sharp = \{x_i = x_i \sqcup f_i^\sharp(x_1, \ldots, x_m) \mid i = 1,2,\ldots,m\}$$

With widening (too coarse)

$$F^\sharp = \{x_i = x_i \nabla f_i^\sharp(x_1, \ldots, x_m) \mid i = 1,2,\ldots,m\}$$

$$x_1 = \alpha(I)$$

$$x_2 = [[\mathrm{x} = 0]]^\sharp x_1 \sqcup [[\mathrm{x} = \mathrm{x} + 2]]^\sharp x_4 \sqcup [[\mathrm{x} = \mathrm{x} + 10]]^\sharp x_5$$

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- It is sufficient to apply the widening on one node in each cycle of the CFG (usually the entry node)

Widening

Where to apply it?

$$F^\sharp = \begin{cases} x_i = x_i \nabla f_i^\sharp(x_1, \ldots, x_m) & \text{if } i \in WP \\ x_i = x_i \sqcup f_i^\sharp(x_1, \ldots, x_m) & \text{otherwise} \end{cases}$$

$$x_1 = \alpha(I)$$

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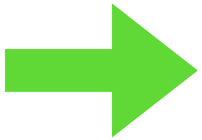
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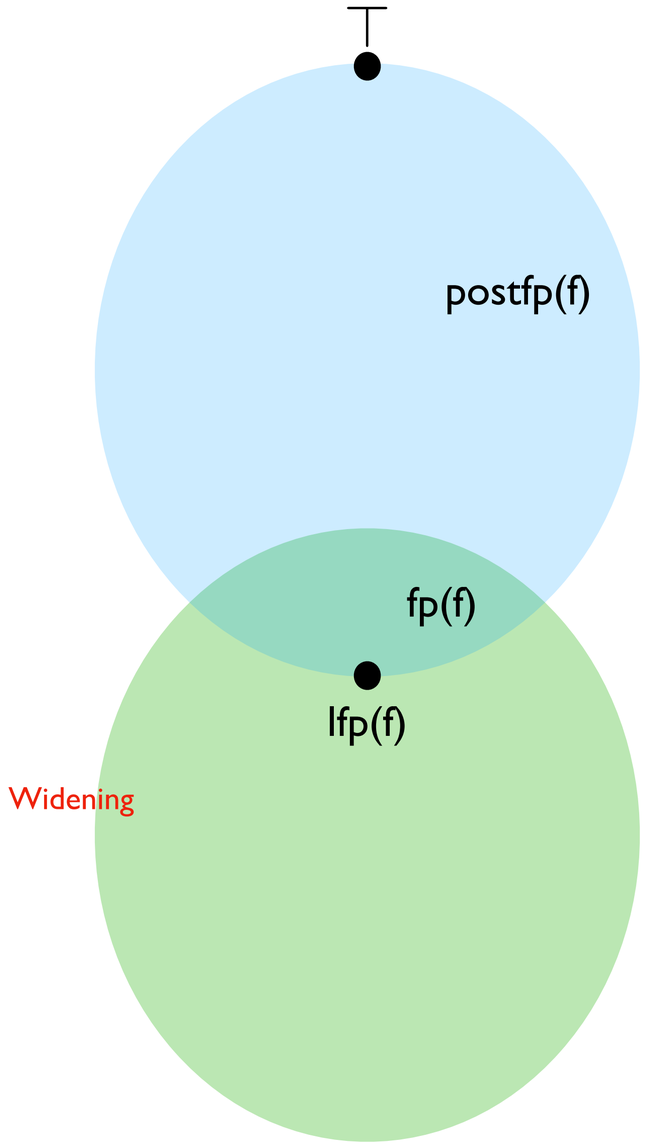


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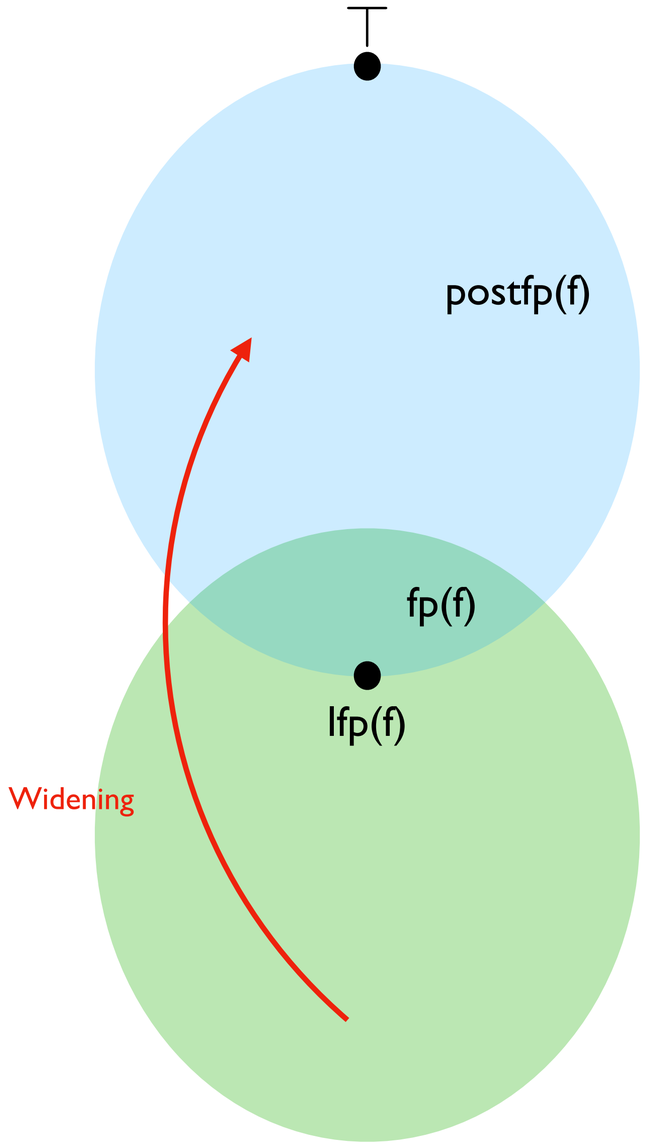
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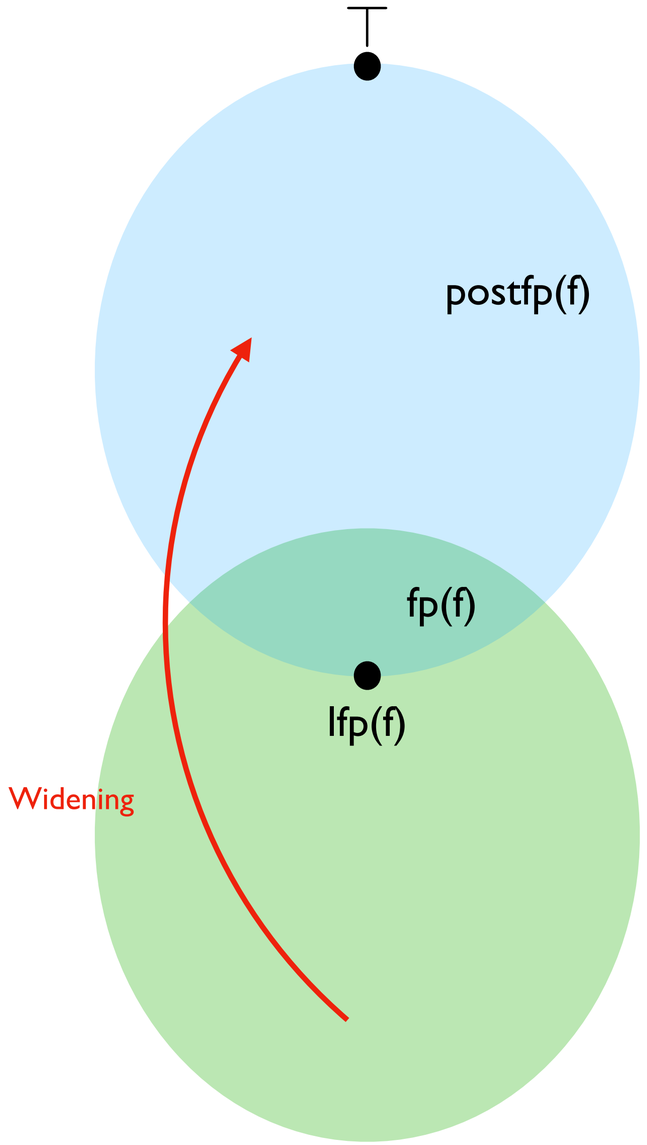
Widening



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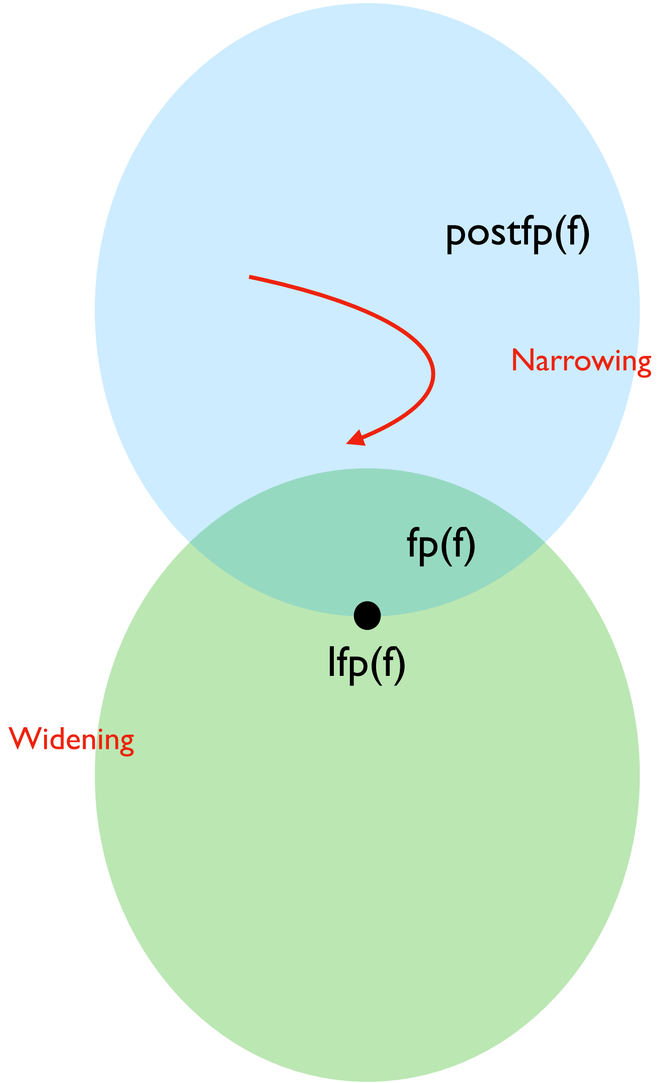


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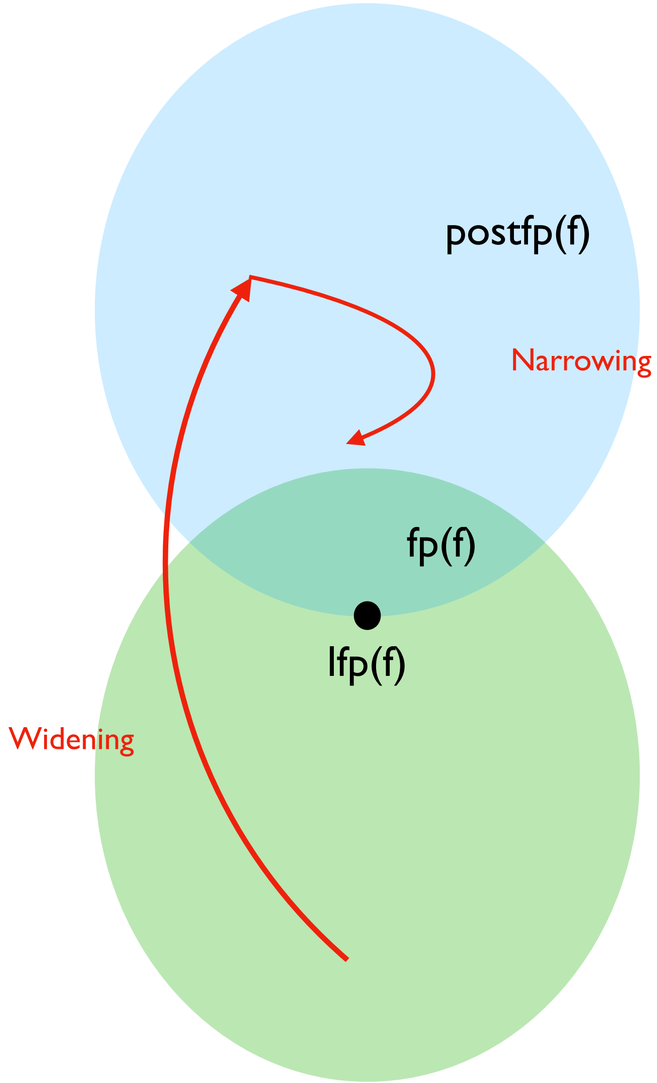


Widening: over-approximates
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Narrowing



Narrowing



Narrowing

- A binary operator $\bigtriangleup : A \times A \to A$ is a narrowing operator in an abstract domain $\langle A, \sqsubseteq \rangle$ if
  - $\forall x, y \in A : x \sqcap y \sqsubseteq x \bigtriangleup y \sqsubseteq x$
  - if ensures convergence: for any sequence $(y^i)_{i \in \mathbb{N}}$ in A, the sequence $(x^i)_{i \in \mathbb{N}}$ computed as $x^0 = y^0, x^{i+1} = x^i \bigtriangleup y^{i+1}$ stabilizes in finite time, namely
    - $\exists k \geq 0. \, x^{k+1} = x^k$

Narrowing for intervals

$$[a,b] \triangle [c,d] = \left[\begin{cases} c & \text{if } a = -\infty \\ a & \text{otherwise} \end{cases}, \begin{cases} d & \text{if } b = +\infty \\ b & \text{otherwise} \end{cases} \right]$$

Narrowing

Where to apply it?

- At the widening points
- starting from the post-fixpoint computed applying widening

$$F^{\sharp} = \begin{cases} x_i = x_i \triangle f_i^{\sharp}(x_1, \ldots, x_m) & \text{if } i \in WP \\ x_i = x_i \sqcap f_i^{\sharp}(x_1, \ldots, x_m) & \text{otherwise} \end{cases}$$

Ascending and descending phases

